Exercises

Introduction: Fibonacci Numbers

1. `memoF` has $\Theta(n)$ entries $\sim$ memory. Can we compute F_n using $O(1)$ memory?

Longest Common Subsequence (LCS) [GT 12.5 , CLRS 15.4]

1. Implement the algorithm described here, both in the recursive memoization version and the bottom-up iterative version.
2. To avoid backtracking, it is possible to maintain a solution for each entry in the table: in addition to the value $\mathsf{L}(i, j)$, store an LCS of $s_1[1 \dots i]$ and $s_2[1 \dots j]$. (i) What is the new runtime? (ii) What would the new size of the table be?

Scheduling Telescope Time [GT 12.3]

1. Write out the pseudocode for the bottom-up iteration version, and analyze the runtime.
2. Compute `p` from sorted `A` in $O(n \log n)$ time.

Knapsack [GT12.6], [CLRS 16.2 (+exercise 35-7)]

1. (not from the slides) **Coin Changing:** Given a supply of coins of denominations `d[1] < d[2] < ... < d[n]`, we wish to make change for a value v; that is, to find a set of coins whose total value is v. This might not be possible: eg, if the denominations are 5 and 10 then we can make change for 15 but not for 12. Give $O(nv)$ time dynamic-programming algorithms for the following problems.
   (a) Each coin denomination may be used an **unlimited number of times**.
   If it is possible to make change for v, output the coins used. Otherwise: output $\textsc{No}$.
   (b) Each coin denomination may be used **at most once**.
   If it is possible to make change for v, output the coins used. Otherwise: output $\textsc{No}$.
   (c) Repeat (a) and (b), but now if it is possible to make change, the output should be the fewest coins possible to do so.

Travelling Salesperson (TSP)

1. Provide an algorithm to backtrack the table.
2. Similar to the LCS case, to compute the table for sets of size j (current column), only need the entries for sets of size $j-1$ (previous column). This reduces the saved table entries, but prevents easy backtracking. So we should store more **info** in the table.
   (i) What is the **info** to be added? (ii) What is the new runtime? (iii) What is the size of the largest such "column" j ?